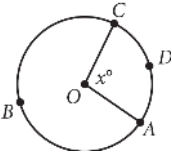


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



The circle above has center O, the length of arc $\widehat{ADC}$ is 5π , and $x = 100$. What is the length of arc $\widehat{ABC}$?

Answer

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

Correct Answer: B

Rationale

Choice B is correct. The ratio of the lengths of two arcs of a circle is equal to the ratio of the measures of the central angles that subtend the arcs. It's given that arc $\widehat{ADC}$ is subtended by a central angle with measure 100° . Since the sum of the measures of the angles about a point is 360° , it follows that arc $\widehat{ABC}$ is subtended by a central angle with measure $360^\circ - 100^\circ = 260^\circ$. If s is the length of arc $\widehat{ABC}$, then s must satisfy the ratio $\frac{s}{5\pi} = \frac{260}{100}$. Reducing the fraction $\frac{260}{100}$ to its simplest form gives $\frac{13}{5}$. Therefore, $\frac{s}{5\pi} = \frac{13}{5}$. Multiplying both sides of $\frac{s}{5\pi} = \frac{13}{5}$ by 5π yields $s = 13\pi$.

Choice A is incorrect. This is the length of an arc consisting of exactly half of the circle, but arc $\widehat{ABC}$ is greater than half of the circle. Choice C is incorrect. This is the total circumference of the circle. Choice D is incorrect. This is half the length of arc $\widehat{ABC}$, not its full length.